

All your memory are belong to ... whom?

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Main screen turn on: free (1)

- ```
  > free -m
                 total        used        free      shared  buff/cache   available
  Mem:           31372       10740        8444         596       13234       20631
  Swap:           2454           0        2454
  ```
- `used + free != total ? (19184 < 31372)` – What Happen ?
- Linux memory management principle: unused memory is wasted memory
  - cache all accessed file contents (and metadata) in memory, as long as free memory exists
  - once free memory is (almost) depleted, the cache is relatively easy to discard
  - *used* tries to say how much is occupied that can't easily be discarded
    - otherwise the number wouldn't be really useful (always close to total)
  ```

Main screen turn on: free (1)

- `> free -m`

```
               total        used        free      shared  buff/cache   available
Mem:           31372       10740        8444         596       13234       20631
Swap:           2454           0        2454
```

- `used + free + buff/cache != total ? (32418 > 31372)`
  - is perhaps part of *buff/cache* considered *used* ?
- `free + buff/cache != available ? (21678 > 20631)`
  - part of *free* or *buff/cache* is not considered *available* ?
- `used + available == total? (31371 ~= 31372)`
  - We get signal!
  - but what is *available* ? and what is *shared* ?

Kernel's accounting: /proc/meminfo

```
MemTotal:       32125696 kB
MemFree:         8627220 kB
MemAvailable:   21107232 kB
Buffers:          258820 kB
Cached:         12902952 kB
SwapCached:            0 kB
Active:          5870672 kB
Inactive:       16277940 kB
Active(anon):       8120 kB
Inactive(anon):  9589756 kB
Active(file):    5862552 kB
Inactive(file):  6688184 kB
Unevictable:          80 kB
Mlocked:              80 kB
SwapTotal:       2513056 kB
SwapFree:        2513056 kB
Zswap:                 0 kB
Zswapped:              0 kB
Dirty:               844 kB
Writeback:             196 kB
AnonPages:        8922944 kB
Mapped:           1683592 kB
Shmem:             611016 kB
KReclaimable:      390056 kB
Slab:              613748 kB
SReclaimable:      390056 kB
SUnreclaim:        223692 kB
KernelStack:        35568 kB
PageTables:         82500 kB
SecPageTables:       3248 kB
NFS_Unstable:          0 kB
Bounce:                0 kB
WritebackTmp:          0 kB
CommitLimit:    18575904 kB
Committed_AS:   21267524 kB
VmallocTotal:   34359738367 kB
VmallocUsed:       84632 kB
VmallocChunk:          0 kB
Percpu:             7680 kB
HardwareCorrupted:     0 kB
AnonHugePages:   1363968 kB
ShmemHugePages:        0 kB
ShmemPmdMapped:        0 kB
FileHugePages:   1062912 kB
FilePmdMapped:    684032 kB
CmaTotal:              0 kB
CmaFree:               0 kB
Unaccepted:            0 kB
HugePages_Total:       0
HugePages_Free:        0
HugePages_Rsvd:        0
HugePages_Surp:        0
Hugepagesize:       2048 kB
Hugetlb:               0 kB
DirectMap4k:      894136 kB
DirectMap2M:    25239552 kB
DirectMap1G:     7340032 kB
```

```
MemTotal:       32125696 kB     Writeback:           196 kB     Percpu:             7680 kB
MemFree:         8627220 kB     AnonPages:        8922944 kB     HardwareCorrupted:     0 kB
MemAvailable:   21107232 kB     Mapped:           1683592 kB     AnonHugePages:   1363968 kB
Buffers:          258820 kB     Shmem:             611016 kB     ShmemHugePages:        0 kB
Cached:         12902952 kB     KReclaimable:      390056 kB     ShmemPmdMapped:        0 kB
SwapCached:            0 kB     Slab:              613748 kB     FileHugePages:   1062912 kB
Active:          5870672 kB     SReclaimable:      390056 kB     FilePmdMapped:    684032 kB
Inactive:       16277940 kB     SUnreclaim:        223692 kB     CmaTotal:              0 kB
Active(anon):       8120 kB     KernelStack:        35568 kB     CmaFree:               0 kB
Inactive(anon):  9589756 kB     PageTables:         82500 kB     Unaccepted:            0 kB
Active(file):    5862552 kB     SecPageTables:       3248 kB     HugePages_Total:       0
Inactive(file):  6688184 kB     NFS_Unstable:           0 kB     HugePages_Free:        0
Unevictable:          80 kB     Bounce:                 0 kB     HugePages_Rsvd:        0
Mlocked:              80 kB     WritebackTmp:           0 kB     HugePages_Surp:        0
SwapTotal:       2513056 kB     CommitLimit:     18575904 kB     Hugepagesize:       2048 kB
SwapFree:        2513056 kB     Committed_AS:    21267524 kB     Hugetlb:               0 kB
Zswap:                 0 kB     VmallocTotal:   34359738367 kB     DirectMap4k:      894136 kB
Zswapped:              0 kB     VmallocUsed:        84632 kB     DirectMap2M:    25239552 kB
Dirty:               844 kB     VmallocChunk:           0 kB     DirectMap1G:     7340032 kB
```

```
MemTotal:       32125696 kB     Writeback:           196 kB     Percpu:             7680 kB
MemFree:         8627220 kB     AnonPages:        8922944 kB     HardwareCorrupted:     0 kB
MemAvailable:   21107232 kB     Mapped:           1683592 kB     AnonHugePages:   1363968 kB
Buffers:          258820 kB     Shmem:             611016 kB     ShmemHugePages:        0 kB
Cached:         12902952 kB     KReclaimable:      390056 kB     ShmemPmdMapped:        0 kB
SwapCached:            0 kB     Slab:              613748 kB     FileHugePages:   1062912 kB
Active:          5870672 kB     SReclaimable:      390056 kB     FilePmdMapped:    684032 kB
Inactive:       16277940 kB     SUnreclaim:        223692 kB     CmaTotal:              0 kB
Active(anon):       8120 kB     KernelStack:        35568 kB     CmaFree:               0 kB
Inactive(anon):  958975                                          Unaccepted:            0 kB
Active(file):    586255                                          HugePages_Total:       0
Inactive(file):  668818                                          HugePages_Free:        0
Unevictable:          8                                          HugePages_Rsvd:        0
Mlocked:              8                                          HugePages_Surp:        0
SwapTotal:       2513056 kB     CommitLimit:       18          kB     Hugepagesize:       2048 kB
SwapFree:        2513056 kB     Committed_AS:   21267524          kB     Hugetlb:               0 kB
Zswap:                 0 kB     VmallocTotal:   34359738367 kB     DirectMap4k:         894136 kB
Zswapped:              0 kB     VmallocUsed:       84632 kB     DirectMap2M:        25239552 kB
Dirty:               844 kB     VmallocChunk:          0 kB     DirectMap1G:         7340032 kB
```

Property of kernel's direct mapping, not exactly occupied memory

```
MemTotal:       32125696 kB
MemFree:         8627220 kB
MemAvailable:   21107232 kB
Buffers:          258820 kB
Cached:         12902952 kB
SwapCached:             0 kB
Active:          587067
Inactive:       1627794
Active(anon):       812
Inactive(anon):  9589756
Active(file):    5862552 kB
Inactive(file):  6688184 kB
Unevictable:          80 kB
Mlocked:              80 kB
SwapTotal:       2513056 kB
SwapFree:        2513056 kB
Zswap:                 0 kB
Zswapped:              0 kB
Dirty:               844 kB
Writeback:             196 kB
AnonPages:         8922944 kB
Mapped:            1683592 kB
Shmem:              611016 kB
KReclaimable:       390056 kB
Slab:
SReclaimable:
SUnreclaim:
KernelStack:
PageTables:
SecPageTables:
NFS_Unstable:          0 kB
Bounce:                0 kB
WritebackTmp:          0 kB
CommitLimit:    18575904 kB
Committed_AS:   21267524 kB
VmallocTotal:   34359738367 kB
VmallocUsed:       84632 kB
VmallocChunk:          0 kB
Percpu:             7680 kB
HardwareCorrupted:     0 kB
AnonHugePages:   1363968 kB
ShmemHugePages:        0 kB
ShmemPmdMapped:        0 kB
FileHugePages:   1062912 kB
FilePmdMapped:    684032 kB
CmaTotal:              0 kB
CmaFree:               0 kB
Unaccepted:            0 kB
HugePages_Total:       0
HugePages_Free:        0
HugePages_Rsvd:        0
HugePages_Surp:        0
Hugepagesize:       2048 kB
Hugetlb:               0 kB
DirectMap4k:      894136 kB
DirectMap2M:    25239552 kB
DirectMap1G:     7340032 kB
```

Hugetlbfs details, beyond the scope here

```
MemTotal:       32125696 kB     Writeback:           196 kB     Percpu:             7680 kB
MemFree:                        ...2944 kB     HardwareCorrupted:     0 kB
MemAvailable:                   ...2 kB     AnonHugePages:   1363968 kB
Buffers:                        ...016 kB     ShmemHugePages:        0 kB
Cached:                         ...056 kB     ShmemPmdMapped:        0 kB
SwapCached:                     ...Slab:        613748 kB     FileHugePages:   1062912 kB
Active:          5870672 kB     SReclaimable:   390056 kB     FilePmdMapped:    684032 kB
Inactive:       16277940 kB     SUnreclaim:     223692 kB     CmaTotal:              0 kB
Active(anon):                   ...kB     CmaFree:               0 kB
Inactive(anon                   ...500 kB     Unaccepted:            0 kB
Active(file):                   ...248 kB     HugePages_Total:       0
Inactive(file)                  ...0     HugePages_Free:        0
Unevictable:           80 kB    Bounce:                 ...B     HugePages_Rsvd:        0
Mlocked:               80 kB    WritebackTmp:           ...kB     HugePages_Surp:        0
SwapTotal:       2510...                                     Hugepagesize:       2048 kB
SwapFree:        2...                                        Hugetlb:               0 kB
Zswap:                                                       DirectMap4k:      894136 kB
Zswapped:                                                    DirectMap2M:    25239552 kB
Dirty:                 844 kB    VmallocChunk:           0 kB     DirectMap1G:     7340032 kB
```

RAM discarded as bad due to HW reported errors (ECC)

Reserved and used by some mobile hardware

Confidential computing guests, temporary after boot

```
MemTotal:       32125696 kB   SwapFree:        2513056 kB   NFS_Unstable:          0 kB
MemFree:         8627220 kB   Zswap:                 0 kB   Bounce:                0 kB
MemAvailable:   21107232 kB   Zswapped:              0 kB   WritebackTmp:          0 kB
Buffers:          258820 kB   Dirty:               844 kB   CommitLimit:    18575904 kB
Cached:         12902952 kB   Writeback:           196 kB   Committed_AS:   21267524 kB
SwapCached:            0 kB   AnonPages:        8922944 kB   VmallocTotal:   34359738367 kB
Active:          5870672 kB   Mapped:           1683592 kB   VmallocUsed:       84632 kB
Inactive:       16277940 kB   Shmem:             611016 kB   VmallocChunk:          0 kB
Active(anon):       8120 kB   KReclaimable:      390056 kB   Percpu:             7680 kB
Inactive(anon):  9589756 kB   Slab:              613748 kB   AnonHugePages:   1363968 kB
Active(file):    5862552 kB   SReclaimable:      390056 kB   ShmemHugePages:        0 kB
Inactive(file):  6688184 kB   SUnreclaim:        223692 kB   ShmemPmdMapped:        0 kB
Unevictable:           80 kB   KernelStack:        35568 kB   FileHugePages:   1062912 kB
Mlocked:               80 kB   PageTables:         82500 kB   FilePmdMapped:    684032 kB
SwapTotal:       2513056 kB   SecPageTables:       3248 kB   Hugetlb:               0 kB
```

```
               total        used        free      shared  buff/cache   available
Mem:           31372       10740        8444         596       13234       20631
Swap:           2454           0        2454
```

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	Dirty:	844 kB	CommitLimit:	18575904 kB
Cached:	12902952 kB	Writeback:	196 kB	Committed_AS:	21267524 kB
SwapCached:	0 kB	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	5870672 kB	Mapped:	1683592 kB	VmallocUsed:	84632 kB
Inactive:	16277940 kB	Shmem:	611016 kB	VmallocChunk:	0 kB
Active(anon):	8120 kB	KReclaimable:	390056 kB	Percpu:	7680 kB
Inactive(anon):	9589756 kB	Slab:	613748 kB	AnonHugePages:	1363968 kB
Active(file):	5862552 kB	SReclaimable:	390056 kB	ShmemHugePages:	0 kB
Inactive(file):	6688184 kB	SUnreclaim:	223692 kB	ShmemPmdMapped:	0 kB
Unevictable:	80 kB	KernelStack:	35568 kB	FileHugePages:	1062912 kB
Mlocked:	80 kB	PageTables:	82500 kB	FilePmdMapped:	684032 kB
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

(free -m)	total	used	free	shared	buff/cache	available
Mem:	31372	10740	8444	596	13234	20631
Swap:	2454	0	2454			

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Legend: Kernel Userspace *Redundant (Subs*

MemTotal:	32125696 kB	SwapFree			
MemFree:	8627220 kB	Zs			
MemAvailable:	21107232 kB	Zswapp			
Buffers:	258820 kB	*Dirty:*			
Cached:	12902952 kB	*Writebac*			
SwapCached:	*0 kB*	AnonPage			kB
Active:	*5870672 kB*	*Mapped:*			
Inactive:	*16277940 kB*	*Shmem:*			
Active(anon):	*8120 kB*	*KReclaimabl*			kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Memory managed by kernel's main page allocator ("buddy allocator") only

Excludes early boot allocations

Changes with memory hotplug/hotremove

Changes with in-kernel guest memory ballooning drivers

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	2588	*irty:*	*844 kB*	CommitLimit:	18575904 kB
		k:	*196 kB*	Committed_AS:	21267524 kB
		s:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Occupied by the compressed userspace data

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	2588	*irty:*	*84* kB	CommitLimit:	18575904 kB
		k:	kB	Committed_AS:	21267524 kB
		s:	892 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapp*		VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	S		VmallocChunk:	0 kB
Active(anon):	*8120 kB*	K		Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	S		AnonHugePages:	*1363968 kB*
Active(file):	*5862552 kB*	*SRec*		ShmemHugePages:	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	ShmemPmdMapped:	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	FileHugePages:	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	FilePmdMapped:	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Occupied by the compressed userspace data

Size of the original uncompressed pages

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*Sh*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 k*		
Unevictable:	*80 kB*	KernelStack:	35568 k		
Mlocked:	*80 kB*	PageTables:	82500 k		
SwapTotal:	2513056 kB	SecPageTables:	3248 kB		

Slab allocator pages
Details in /proc/slabinfo
Also large kmalloc()

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*Sh[illegible]ages:*	*0 kB*
[illegible](file):	*[illegible] kB*	*SUnreclaim:*	*223692 k*		
		[illegible]ernelStack:	35568 k		
		[illegible]ageTables:	82500 k		
		[illegible]ecPageTables:	3248 kB		

Slab caches with shrinkers that can discard objects when asked

Slab allocator pages
Details in `/proc/slabinfo`
Also large kmalloc()

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*95[illegible] kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
		SReclaimable:	*390056 kB*	*ShmemHugePages:*	*0 kB*
		SUnreclaim:	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
		KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
		PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
		SecPageTables:	3248 kB	Hugetlb:	0 kB

Used to be SReclaimable plus other users – ION allocator, now gone

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
		ped:	*1683592 kB*	VmallocUsed:	84632 kB
		em:	*611016 kB*	VmallocChunk:	0 kB
		claimable:	*390056 kB*	Percpu:	7680 kB
		b:	613748 kB	*AnonHugePages:*	*1363968 kB*
		claimable:	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*60*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Kernel stacks of kernel threads (kthreads) But all userspace threads also have them

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal: 32125696 kB	SwapFree: 2513056 kB	NFS_Unstable: 0 kB
MemFree: 8627220 kB	Zswap: 0 kB	Bounce: 0 kB
MemAvailable: 21107232 kB	Zswapped: 0 kB	WritebackTmp: 0 kB
Buffers: 258820 kB	*Dirty: 844 kB*	CommitLimit: 18575904 kB
Cached: 12902952 kB	*Writeback: 196 kB*	Committed_AS: 21267524 kB
SwapCached: 0 kB	AnonPages: 8922944 kB	VmallocTotal: 34359738367 kB
Active: 5870672 kB	*Mapped: 1683592 kB*	VmallocUsed: 84632 kB
Inactive: 16277940 kB	*Shmem: 611016 kB*	VmallocChunk: 0 kB
	claimable: 390056 kB	Percpu: 7680 kB
	b: 613748 kB	*AnonHugePages: 1363968 kB*
	claimable: 390056 kB	*ShmemHugePages: 0 kB*
Inactive(file): 6	*SUnreclaim: 223692 kB*	*ShmemPmdMapped: 0 kB*
Unevictable: 80	KernelStack: 35568 kB	*FileHugePages: 1062912 kB*
Mlocked: 80 kB	PageTables: 82500 kB	*FilePmdMapped: 684032 kB*
SwapTotal: 2513056 kB	SecPageTables: 3248 kB	Hugetlb: 0 kB

Page tables for userspace processes

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*		
		*cl*aimable:	*390056 k*		
		b:	613748 k		
		*cl*aimable:	*390056 kB*	S Pages:	*0 kB*
Inactive(file):	*6*	*SUnreclaim:*	*223692 kB*	mdMapped:	*0 kB*
Unevictable:	*80*	KernelStack:	35568 kB	eHugePages:	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Page tables for userspace processes

Secondary page tables (IOMMU, KVM)

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

Historical, hardcoded 0

MemTotal:				NFS_Unstable:	0 kB
MemFree:			0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:				NFS_Unstable:	0 kB
MemFree:			0 kB	Bounce:	0 kB
MemAvailable:			kB	WritebackTmp:	0 kB
Buffers:			*844 kB*	CommitLimit:	18575904 kB
Cached:			*196 kB*	Committed_AS:	21267524 kB
SwapCached:			8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Historical, hardcoded 0

Bounce I/O buffers (temporary, when zero-copy not possible)

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:				NFS_Unstable:	0 kB
MemFree:			0 kB	Bounce:	0 kB
MemAvailable:			kB	WritebackTmp:	0 kB
Buffers:			*844 kB*	CommitLimit:	18575904 kB
Cached:			*19*	Committed_AS:	21267524 kB
SwapCached:			*8* kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*			VmallocUsed:	84632 kB
Inactive:	*16277940 kB*			VmallocChunk:	0 kB
Active(anon):	*8120 kB*			Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Historical, hardcoded 0

Bounce I/O buffers (temporary, when zero-copy not possible)

Temporary writeback buffer (only FUSE uses)

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
			168350 kB	VmallocUsed:	84632 kB
			6 kB	VmallocChunk:	0 kB
			056 kB	Percpu:	7680 kB
			13748 kB	*AnonHugePages:*	*1363968 kB*
			390056 kB	*ShmemHugePages:*	*0 kB*
			223692 kB	*ShmemPmdMapped:*	*0 kB*
			35568 kB	*FileHugePages:*	*1062912 kB*
			82500 kB	*FilePmdMapped:*	*684032 kB*
			3248 kB	Hugetlb:	0 kB

Sum of sizes of *accountable* mmap( ) areas of all processes (private, writable, not hugetlb)

- where writes by the processes can populate pages that are not pure page cache (AS = address spaces?)

A limit for `Committed_AS` tunable by sysctl, defaults to half the RAM size plus all swap

Only applies to `OVERCOMMIT_NEVER` (`vm.overcommit_memory = 2`)

Sum of sizes of *accountable* `mmap()` areas of all processes (private, writable, not hugetlb)

- where writes by the processes can populate pages that are not pure page cache

(AS = address spaces?)

```
other) Not consumed memory
513056 kB    NFS_Unstable:          0 kB
      0 kB   Bounce:                0 kB
      0 kB   WritebackTmp:          0 kB
             CommitLimit:    18575904 kB
 196 kB      Committed_AS:   21267524 kB
8922944 kB   VmallocTotal:   34359738367 kB
168359 kB    VmallocUsed:       84632 kB
  6 kB       VmallocChunk:          0 kB
 056 kB      Percpu:             7680 kB
13748 kB     AnonHugePages:   1363968 kB
390056 kB    ShmemHugePages:        0 kB
223692 kB    ShmemPmdMapped:        0 kB
 35568 kB    FileHugePages:   1062912 kB
 82500 kB    FilePmdMapped:    684032 kB
  3248 kB    Hugetlb:               0 kB
```

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Legend: Kernel Userspace *Redundant (Subset of another)* Not consumed memory

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	8627220 kB	Zswap:	0 kB	Bounce:	0 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	WritebackTmp:	0 kB
Buffers:	258820 kB	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	12902952 kB	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	VmallocUsed:	84632 kB
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	Percpu:	7680 kB
Inactive(anon):	*9589756 kB*	Slab:	613748 kB	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	KernelStack:	35568 kB	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	PageTables:	82500 kB	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Size of virtual region in the kernel address space where vmalloc() areas are placed (rather useless number…)

L				Not consumed memory	
M			kB	NFS_Unstable:	0 kB
M			kB	Bounce:	0 kB
M			kB	WritebackTmp:	0 kB
Buffers:	238820 kB	Dirty:	844 kB	CommitLimit:	18575904 kB
Cached:	12902952 kB	Writeback:	06 kB	Committed_AS:	21267524 kB
SwapCached:	0 kB	AnonPages:	8922944 kB	VmallocTotal:	34359738367 kB
Active:	5870672 kB	Mapped:	1683592 kB	VmallocUsed:	84632 kB
Inactive:	16277940 kB	Shmem:	611016 kB	VmallocChunk:	0 kB
Active(anon):	8120 kB	KReclaimable:	390056 kB	Percpu:	7680 kB
Inactive(anon):	9589756 kB	Slab:	613748 kB	AnonHugePages:	1363968 kB
Active(file):	5862552 kB	SReclaimable:	390056 kB	ShmemHugePages:	0 kB
Inactive(file):	6688184 kB	SUnreclaim:	223692 kB	ShmemPmdMapped:	0 kB
Unevictable:	80 kB	KernelStack:	35568 kB	FileHugePages:	1062912 kB
Mlocked:	80 kB	PageTables:	82500 kB	FilePmdMapped:	684032 kB
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Size of virtual region in the kernel address space where vmalloc() areas are placed (rather useless number…)

Amount of memory backing the actually used vmalloc() areas

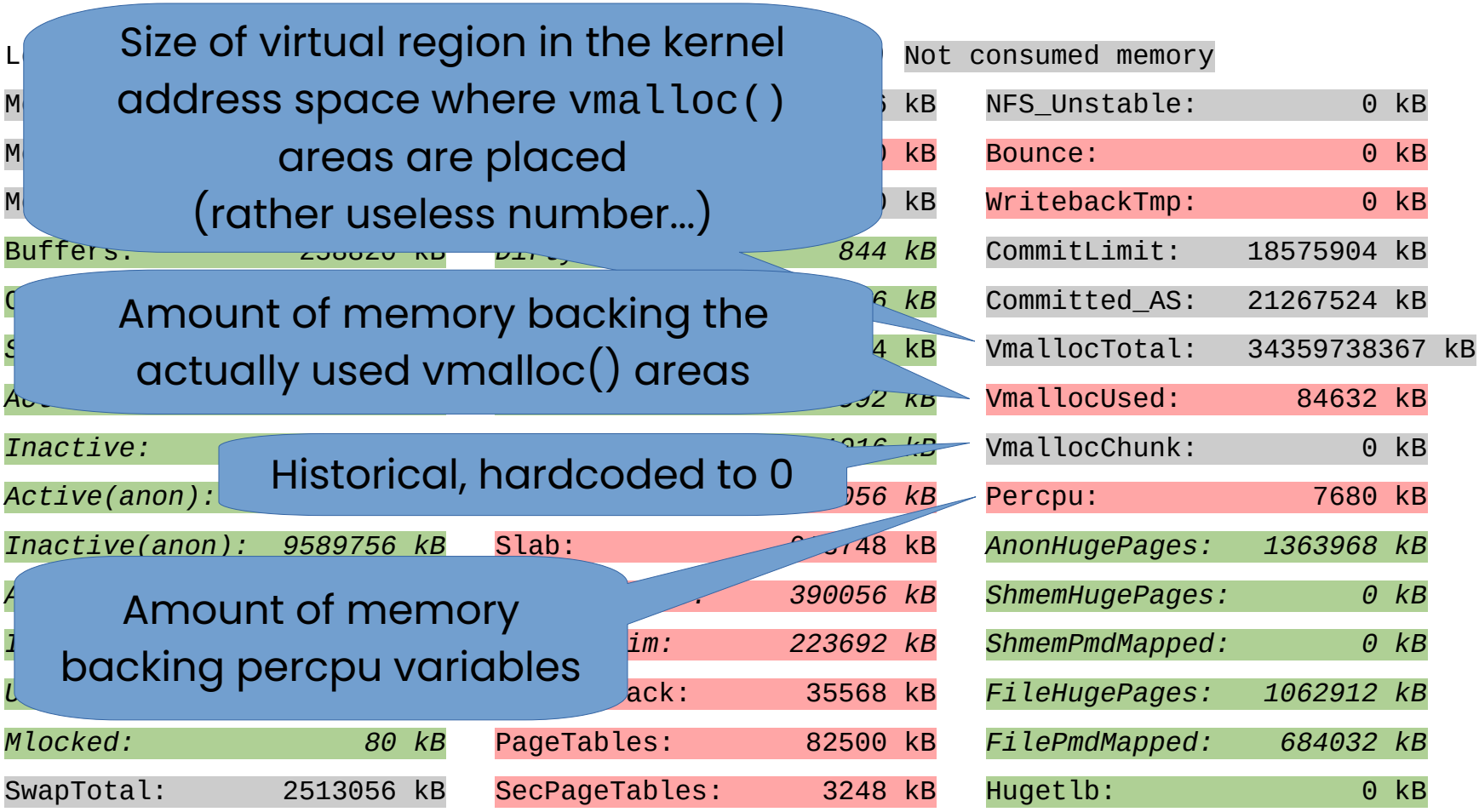
L				Not consumed memory	
M			kB	NFS_Unstable:	0 kB
M			kB	Bounce:	0 kB
M			kB	WritebackTmp:	0 kB
Buffers:	238820 kB	Dirty:	844 kB	CommitLimit:	18575904 kB
C			6 kB	Committed_AS:	21267524 kB
S			4 kB	VmallocTotal:	34359738367 kB
Ac			92 kB	VmallocUsed:	84632 kB
Inactive:	16277940 kB	Shmem:	611016 kB	VmallocChunk:	0 kB
Active(anon):	8120 kB	KReclaimable:	390056 kB	Percpu:	7680 kB
Inactive(anon):	9589756 kB	Slab:	613748 kB	AnonHugePages:	1363968 kB
Active(file):	5862552 kB	SReclaimable:	390056 kB	ShmemHugePages:	0 kB
Inactive(file):	6688184 kB	SUnreclaim:	223692 kB	ShmemPmdMapped:	0 kB
Unevictable:	80 kB	KernelStack:	35568 kB	FileHugePages:	1062912 kB
Mlocked:	80 kB	PageTables:	82500 kB	FilePmdMapped:	684032 kB
SwapTotal:	2513056 kB	SecPageTables:	3248 kB	Hugetlb:	0 kB

Size of virtual region in the kernel address space where vmalloc() areas are placed (rather useless number...)

Amount of memory backing the actually used vmalloc() areas

Historical, hardcoded to 0

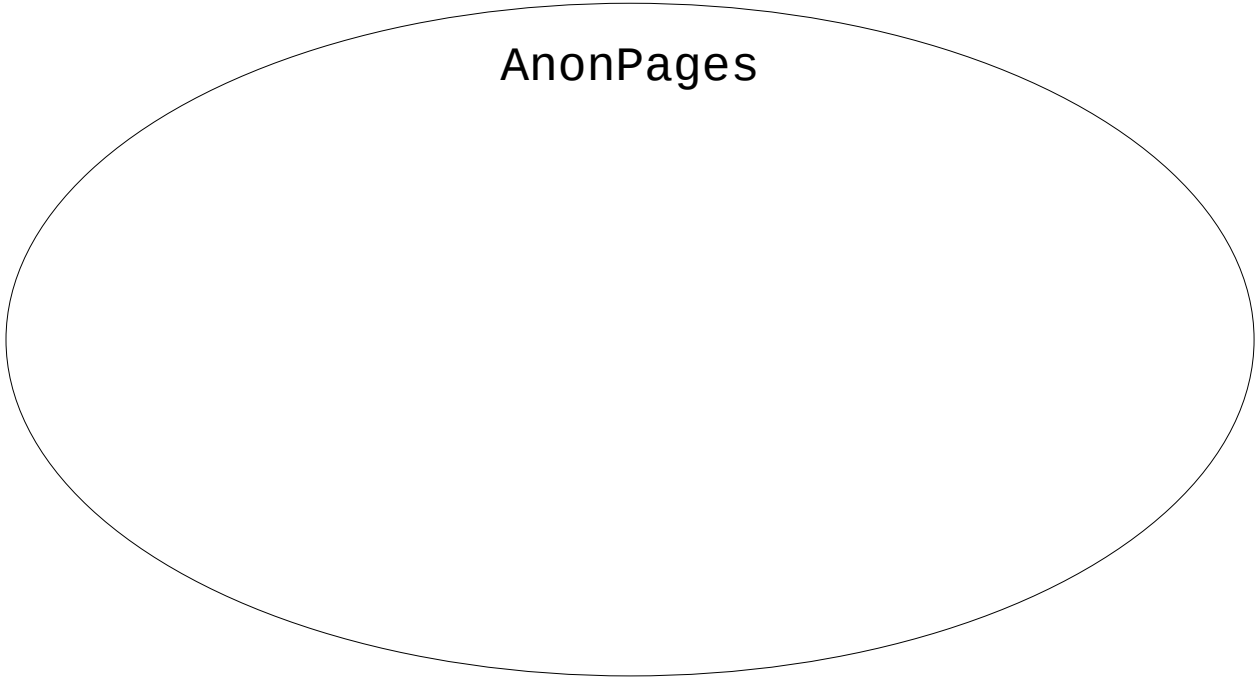
```
L                                                    Not consumed memory
M                                                 kB  NFS_Unstable:          0 kB
M                                                 kB  Bounce:                0 kB
M                                                 kB  WritebackTmp:          0 kB
Buffers:          258820 kB  Dirty:               844 kB  CommitLimit:     18575904 kB
C                                                 kB  Committed_AS:    21267524 kB
S                                                 kB  VmallocTotal:   34359738367 kB
Ac                                                kB  VmallocUsed:        84632 kB
Inactive:                                         kB  VmallocChunk:          0 kB
Active(anon):                                     kB  Percpu:               7680 kB
Inactive(anon):  9589756 kB  Slab:           613748 kB  AnonHugePages:   1363968 kB
Active(file):    5862552 kB  SReclaimable:   390056 kB  ShmemHugePages:        0 kB
Inactive(file):  6688184 kB  SUnreclaim:     223692 kB  ShmemPmdMapped:        0 kB
Unevictable:          80 kB  KernelStack:     35568 kB  FileHugePages:   1062912 kB
Mlocked:              80 kB  PageTables:      82500 kB  FilePmdMapped:    684032 kB
SwapTotal:       2513056 kB  SecPageTables:    3248 kB  Hugetlb:               0 kB
```


Size of virtual region in the kernel address space where vmalloc() areas are placed (rather useless number...)
Amount of memory backing the actually used vmalloc() areas
Historical, hardcoded to 0
Amount of memory backing percpu variables
Not consumed memory
NFS_Unstable: 0 kB
Bounce: 0 kB
WritebackTmp: 0 kB
CommitLimit: 18575904 kB
Committed_AS: 21267524 kB
VmallocTotal: 34359738367 kB
VmallocUsed: 84632 kB
VmallocChunk: 0 kB
Percpu: 7680 kB
AnonHugePages: 1363968 kB
ShmemHugePages: 0 kB
ShmemPmdMapped: 0 kB
FileHugePages: 1062912 kB
FilePmdMapped: 684032 kB
Hugetlb: 0 kB
Inactive:
Active(anon):
Inactive(anon): 9589756 kB
Slab:
Mlocked: 80 kB
PageTables: 82500 kB
SwapTotal: 2513056 kB
SecPageTables: 3248 kB

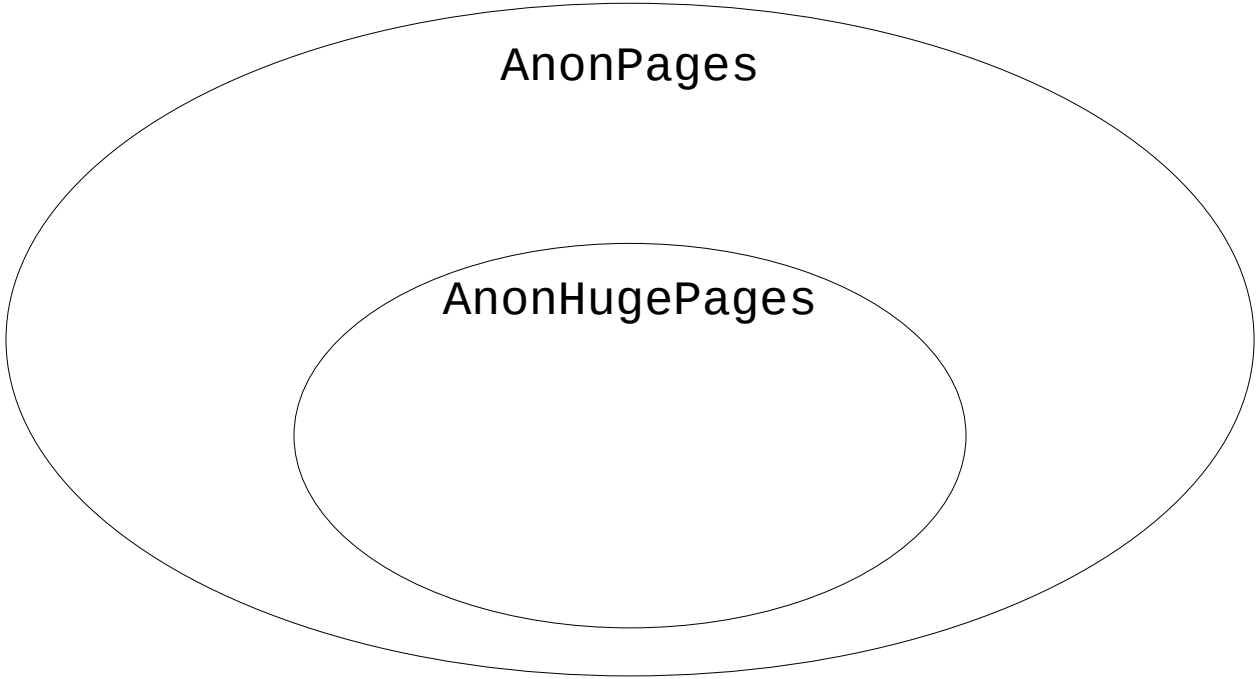
Anonymous memory vs page cache

- Anonymous pages – not backed by files
  - allocated mainly by faulting in `mmap(MAP_PRIVATE | MAP_ANONYMOUS)` areas
  - lifetime tied to the process(es) mapping them
    - can effectively become shared during `fork()`, due to copy-on-write (COW) semantics
  - cannot be simply written back to a file and discarded, only swapped out to swap
- Page cache – pages backed by files on a FS, or a block device
  - allocated by accessing files that are `mmap()`ed, or read/written to in other ways
  - lifetime independent of processes, can be discarded once not dirty…
- Shared memory (shmem, tmpfs) – a hybrid between anonymous and page cache
  - a RAM only filesystem (`/dev/shm`), also used for `shmget()` and `mmap(MAP_SHARED | MAP_ANONYMOUS)`
  - behaves a lot like page cache – lifetime independent of processes in most cases
  - but there are no persistent files backing it – can be swapped out like anonymous pages

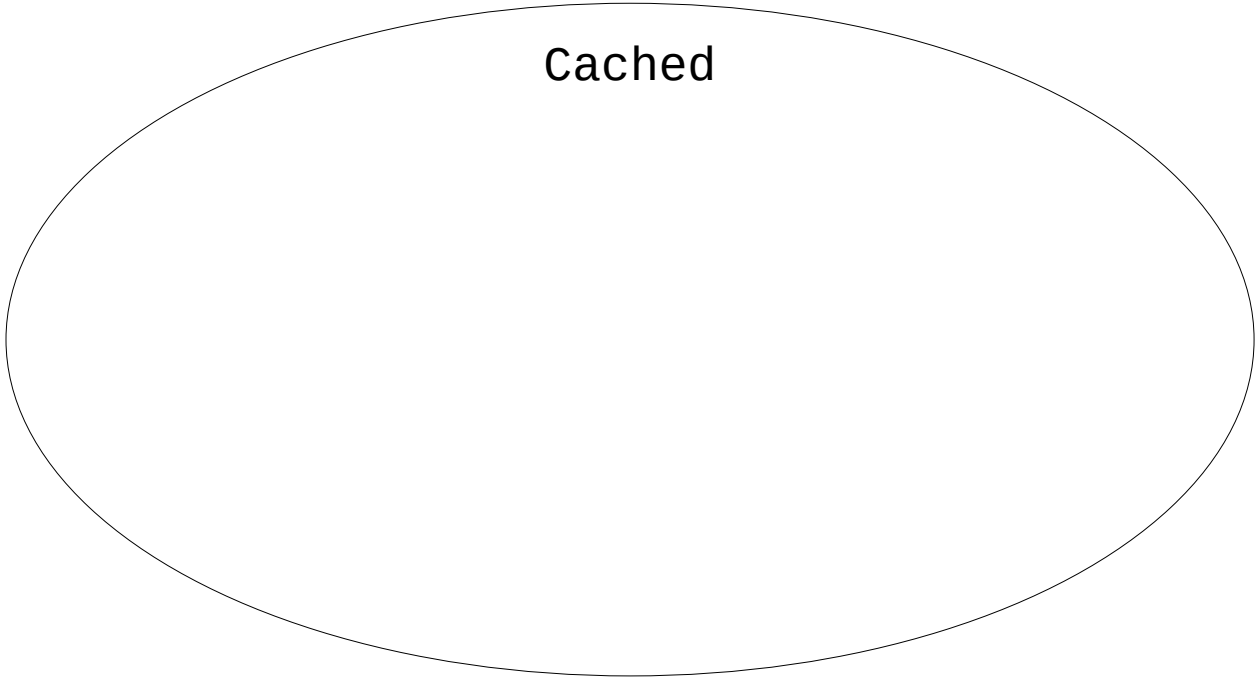
Anonymous counters in detail



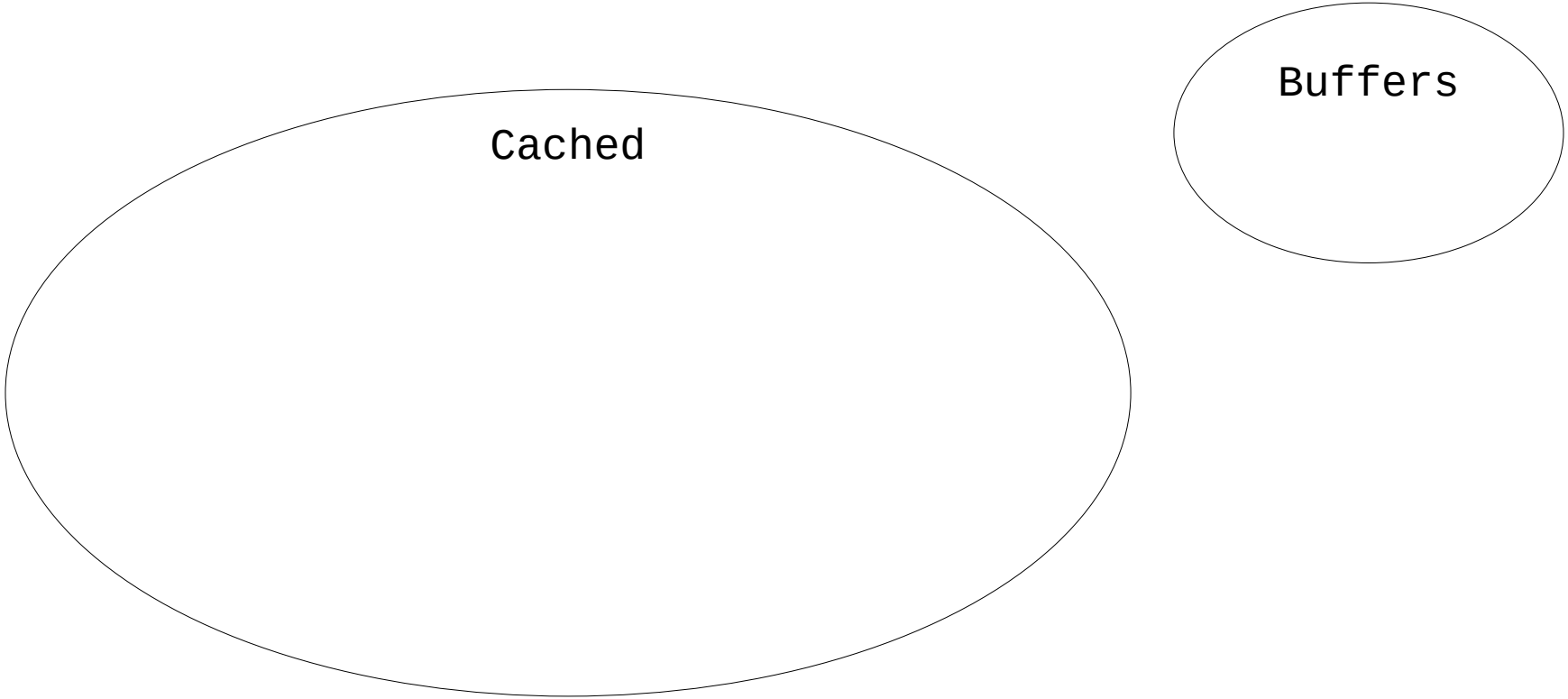
Anonymous counters in detail



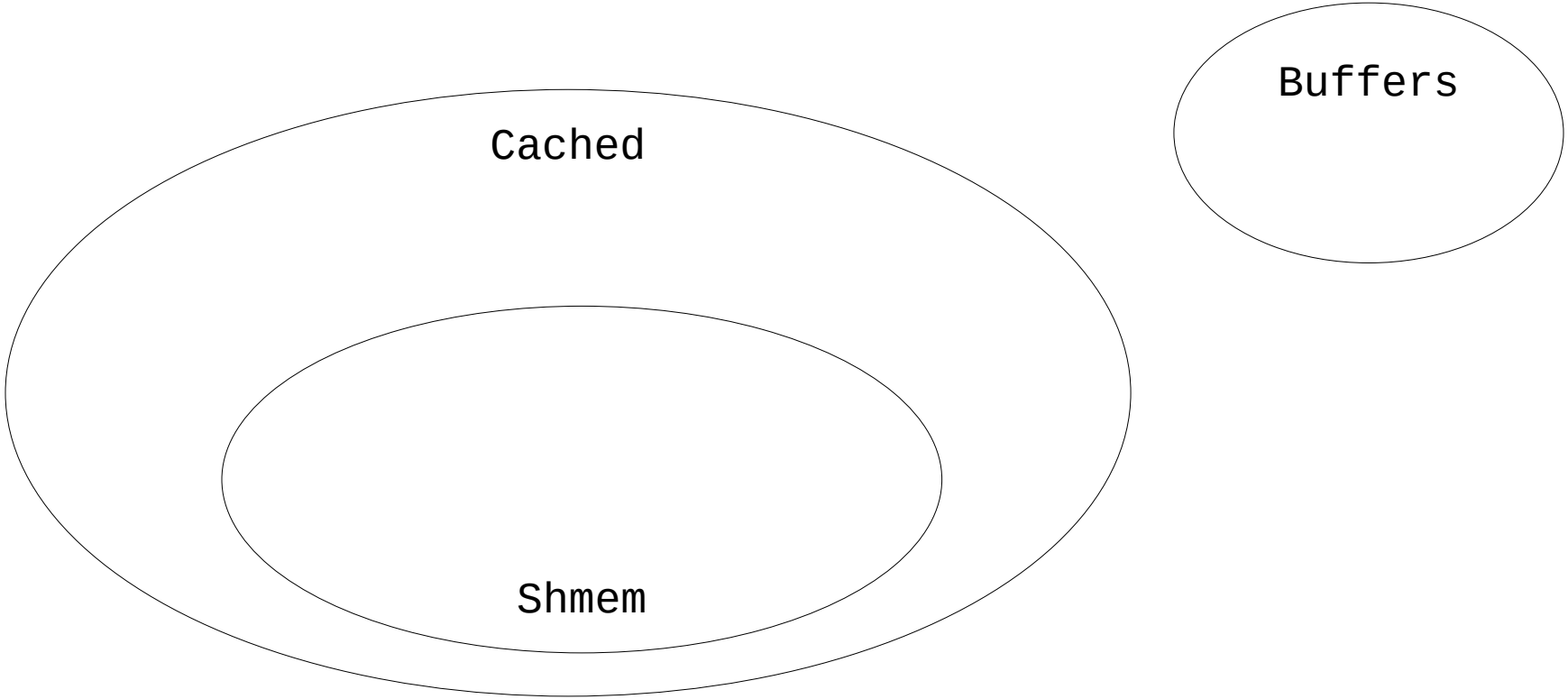
Page cache counters in detail



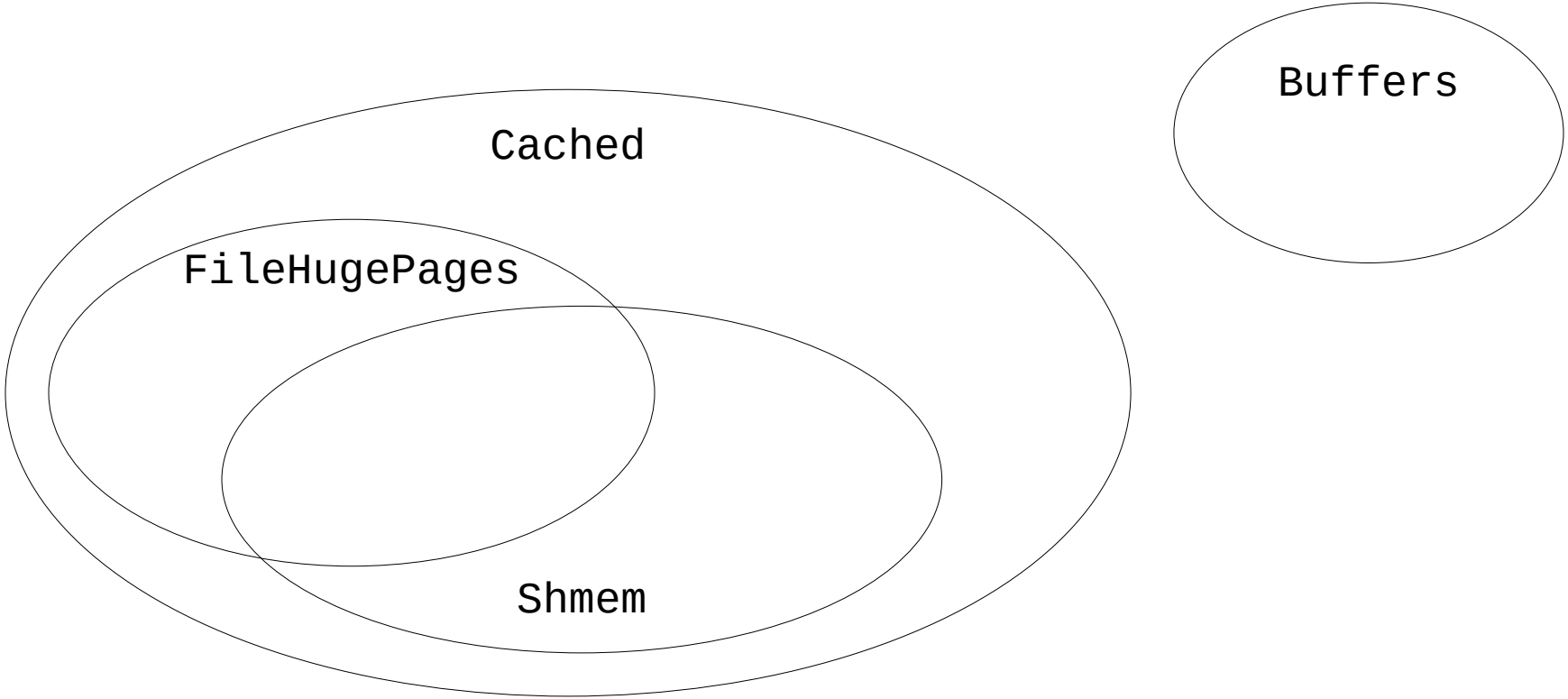
Page cache counters in detail



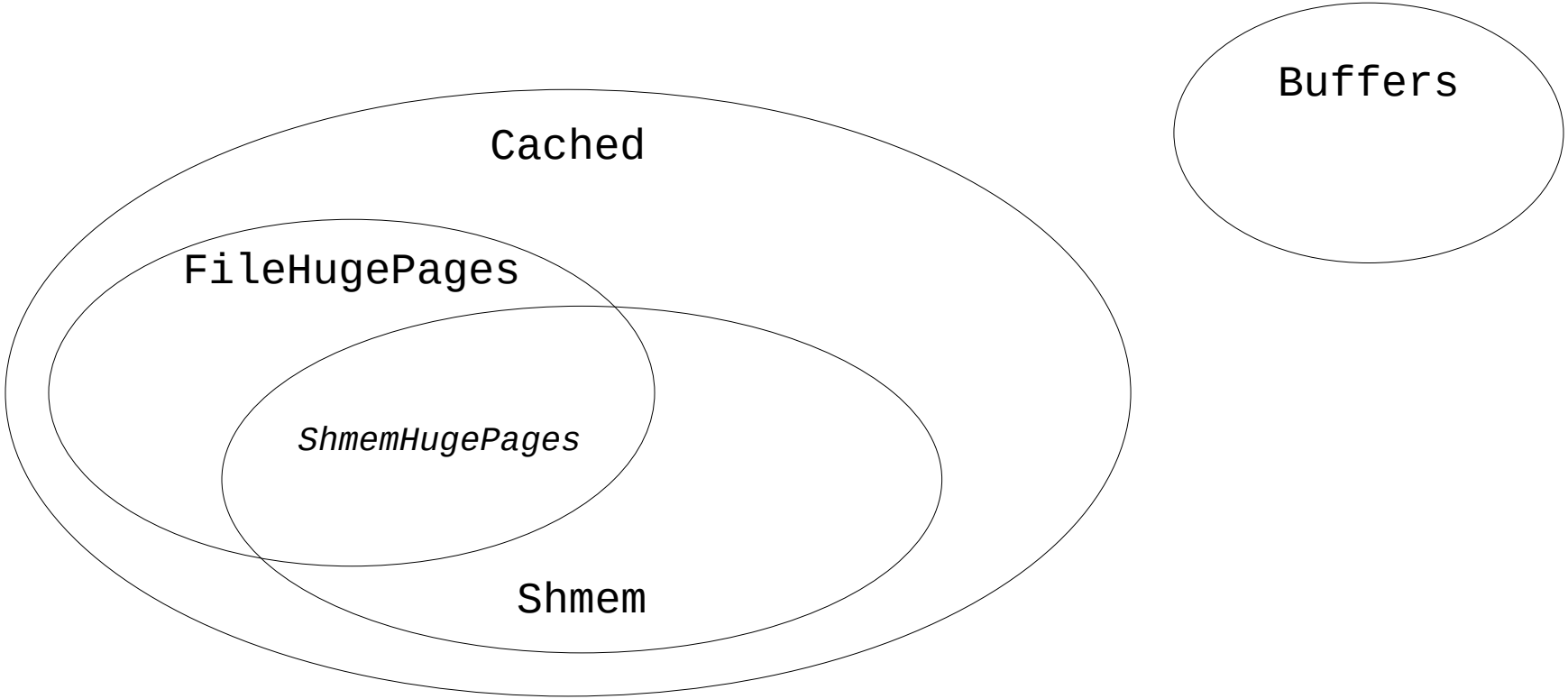
Page cache counters in detail



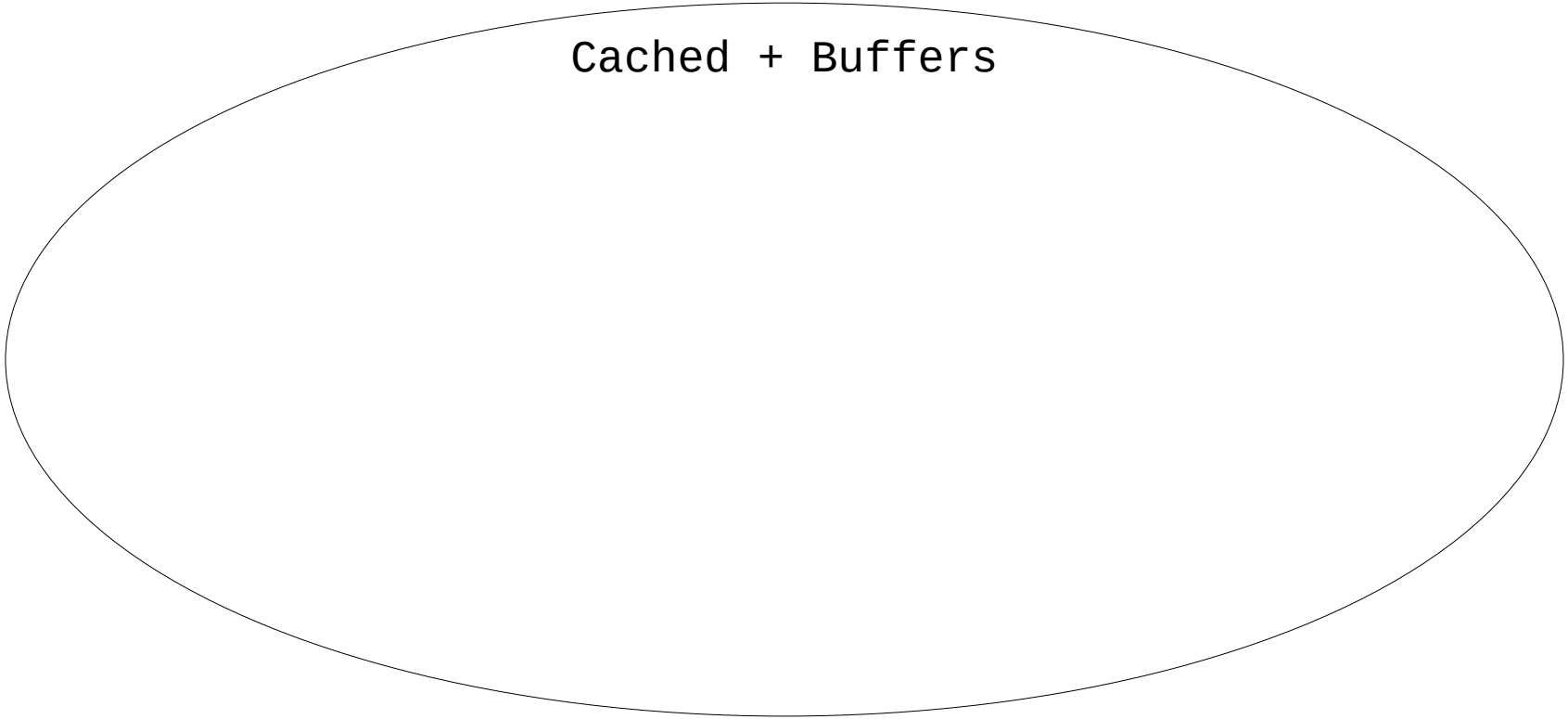
Page cache counters in detail



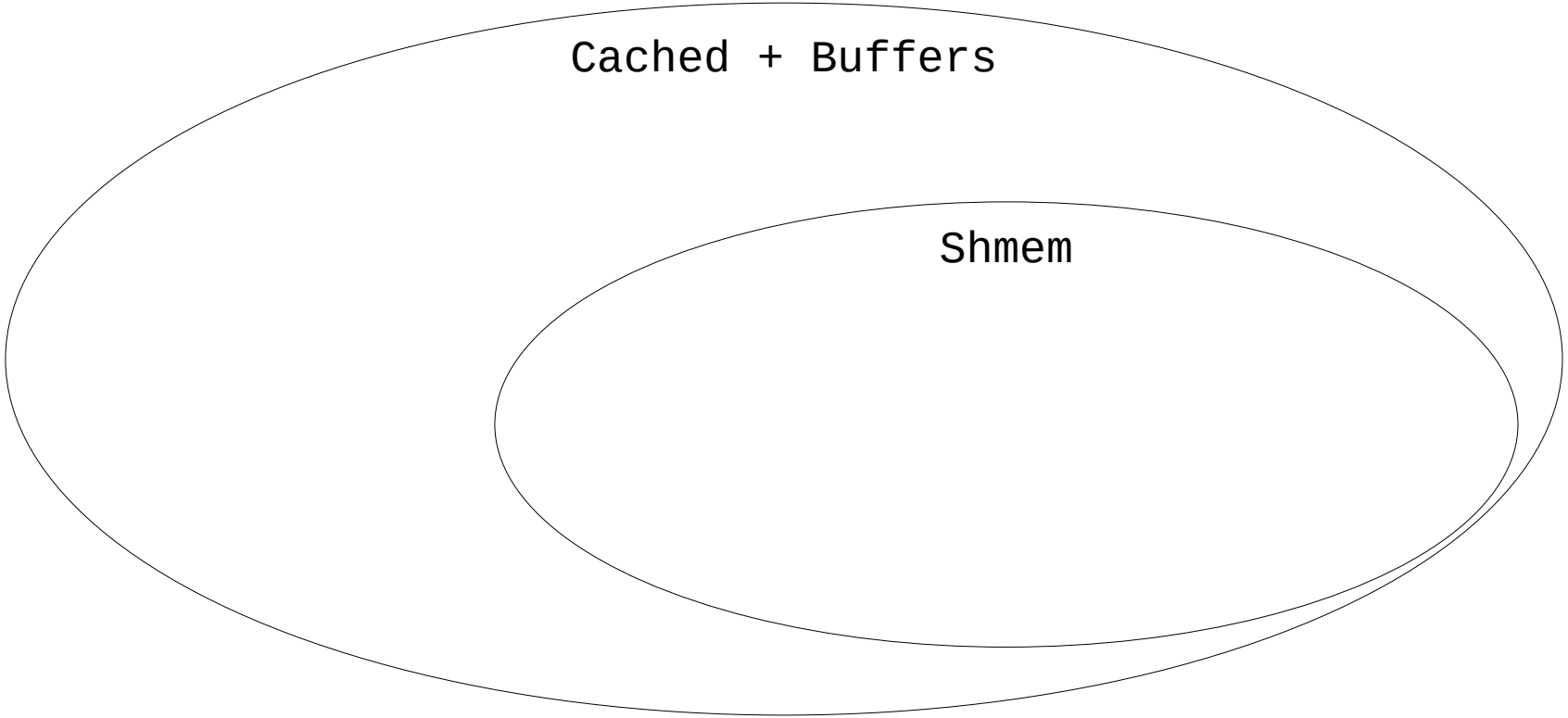
Page cache counters in detail



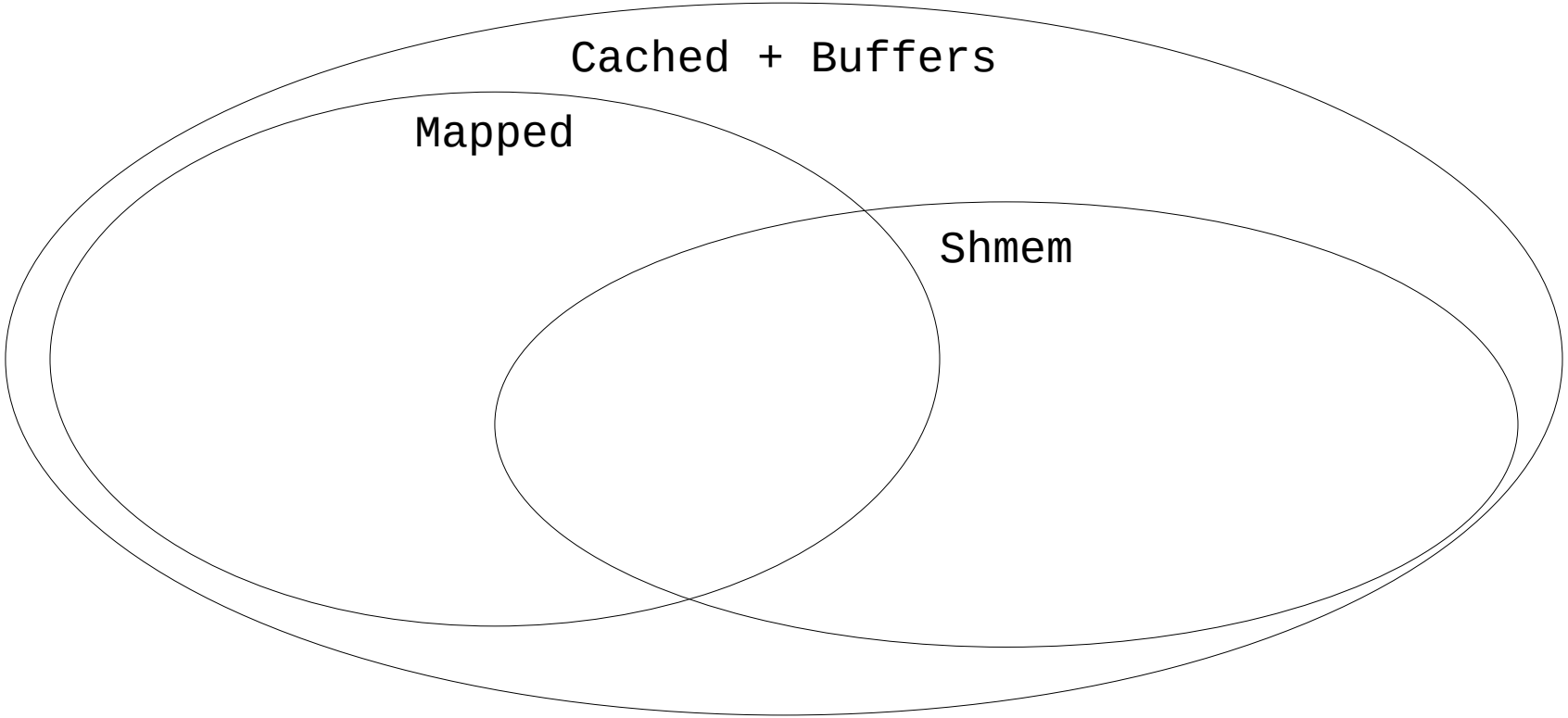
Page cache counters in detail – (Pmd)Mapped



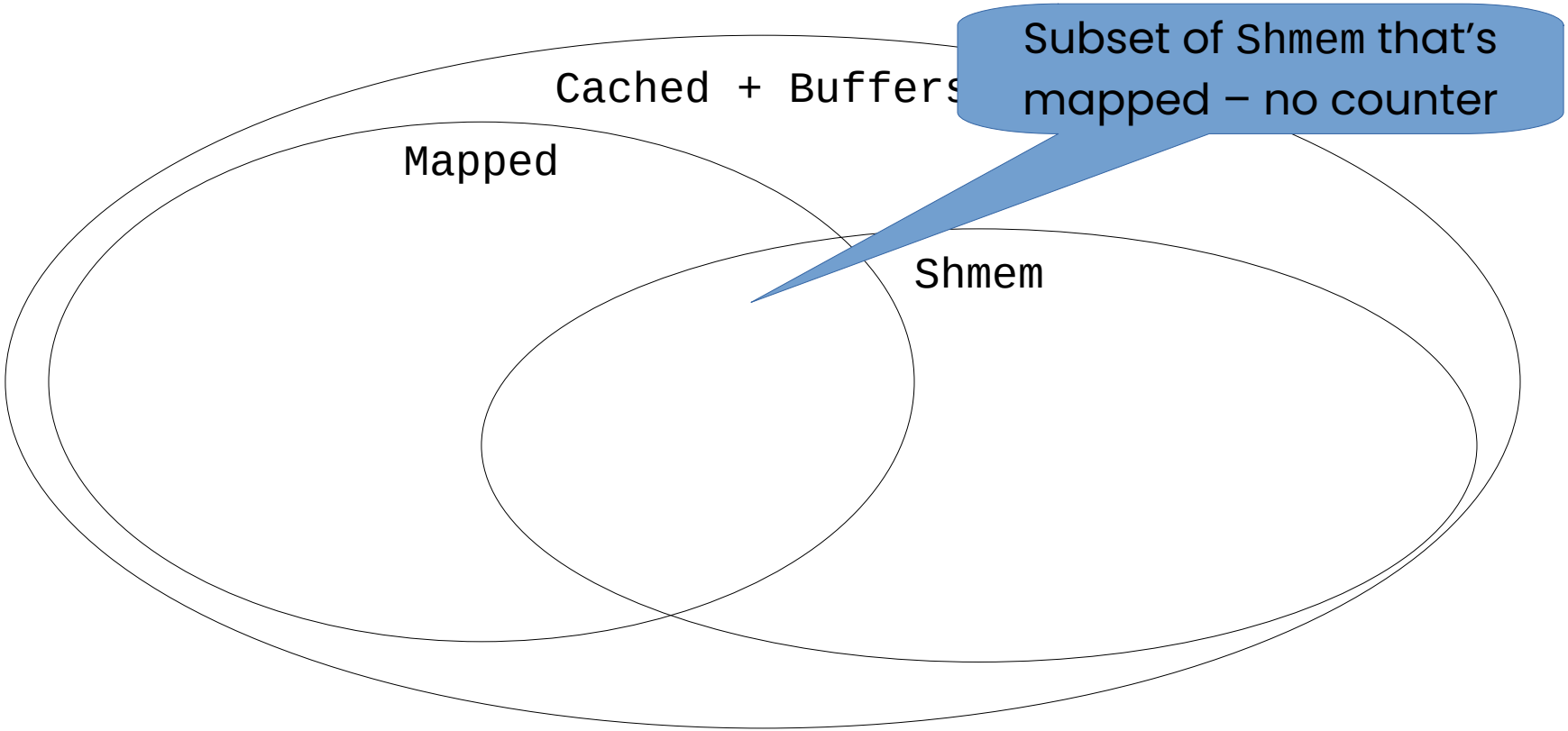
Page cache counters in detail – (Pmd)Mapped



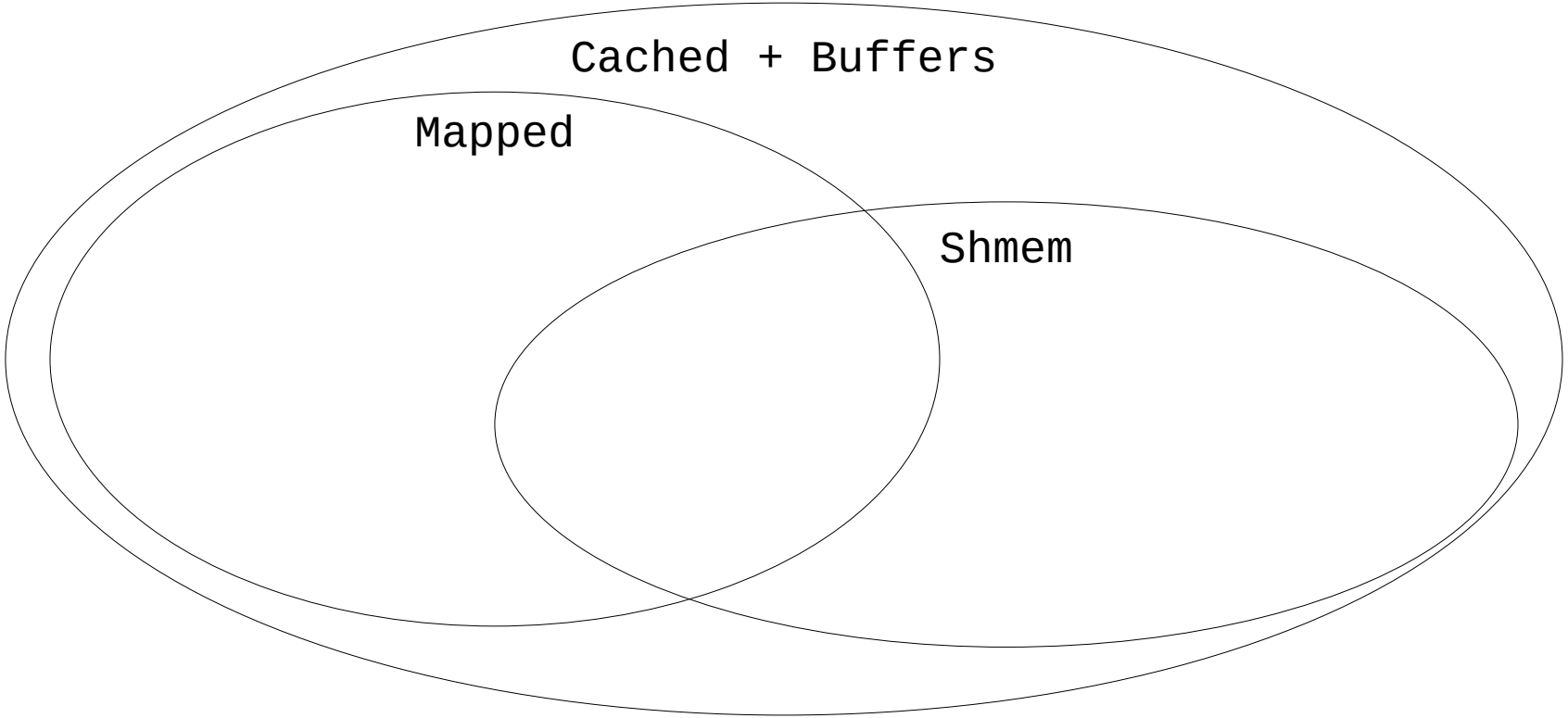
Page cache counters in detail – (Pmd)Mapped



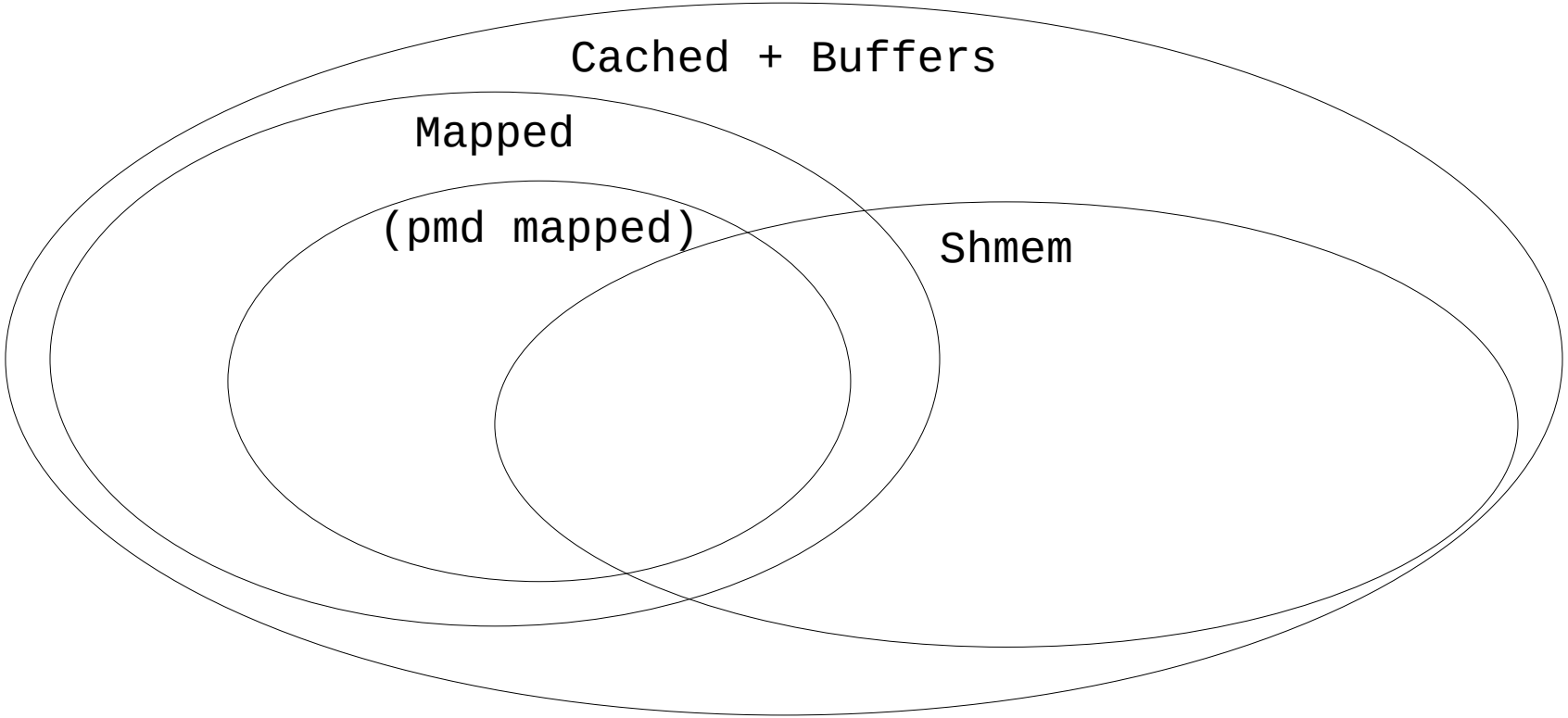
Page cache counters in detail – (Pmd)Mapped



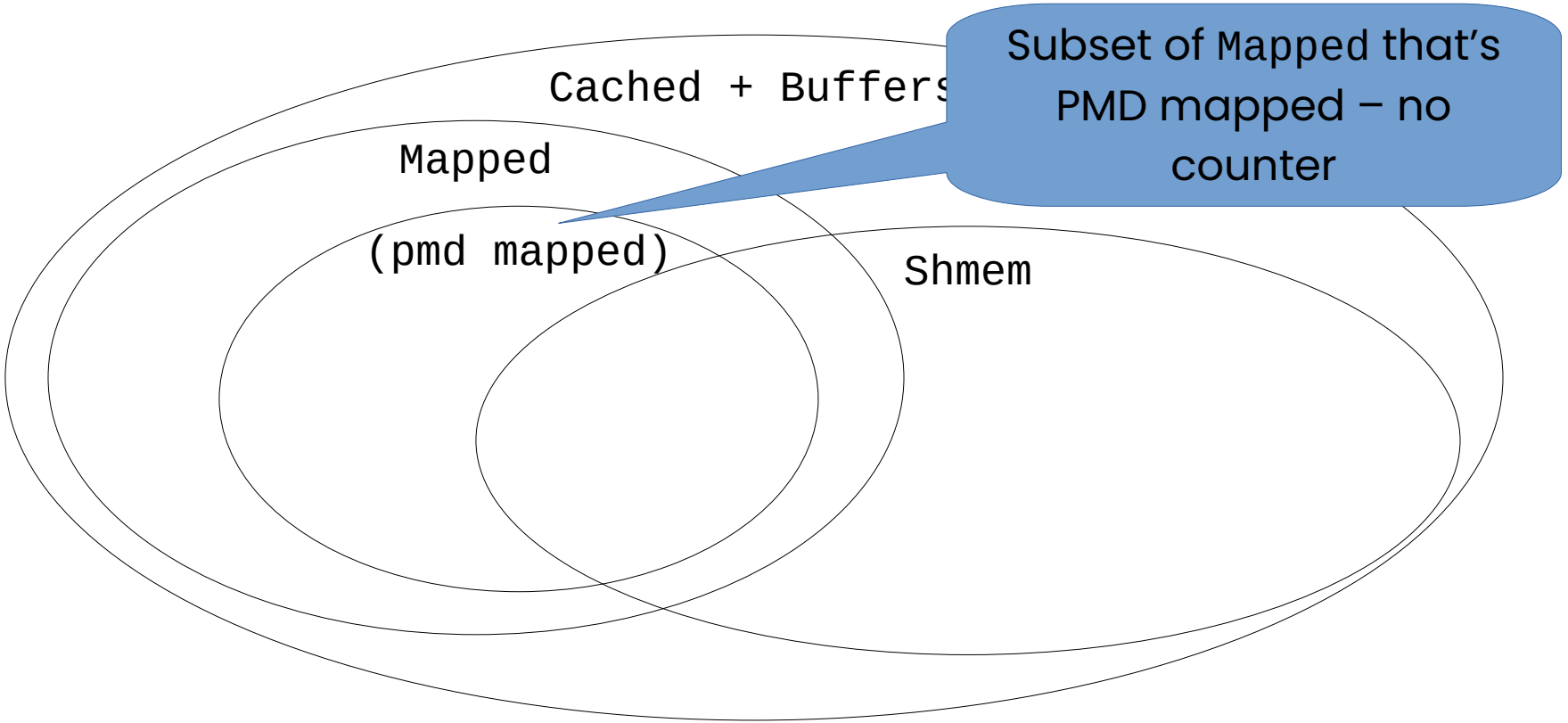
Page cache counters in detail – (Pmd)Mapped



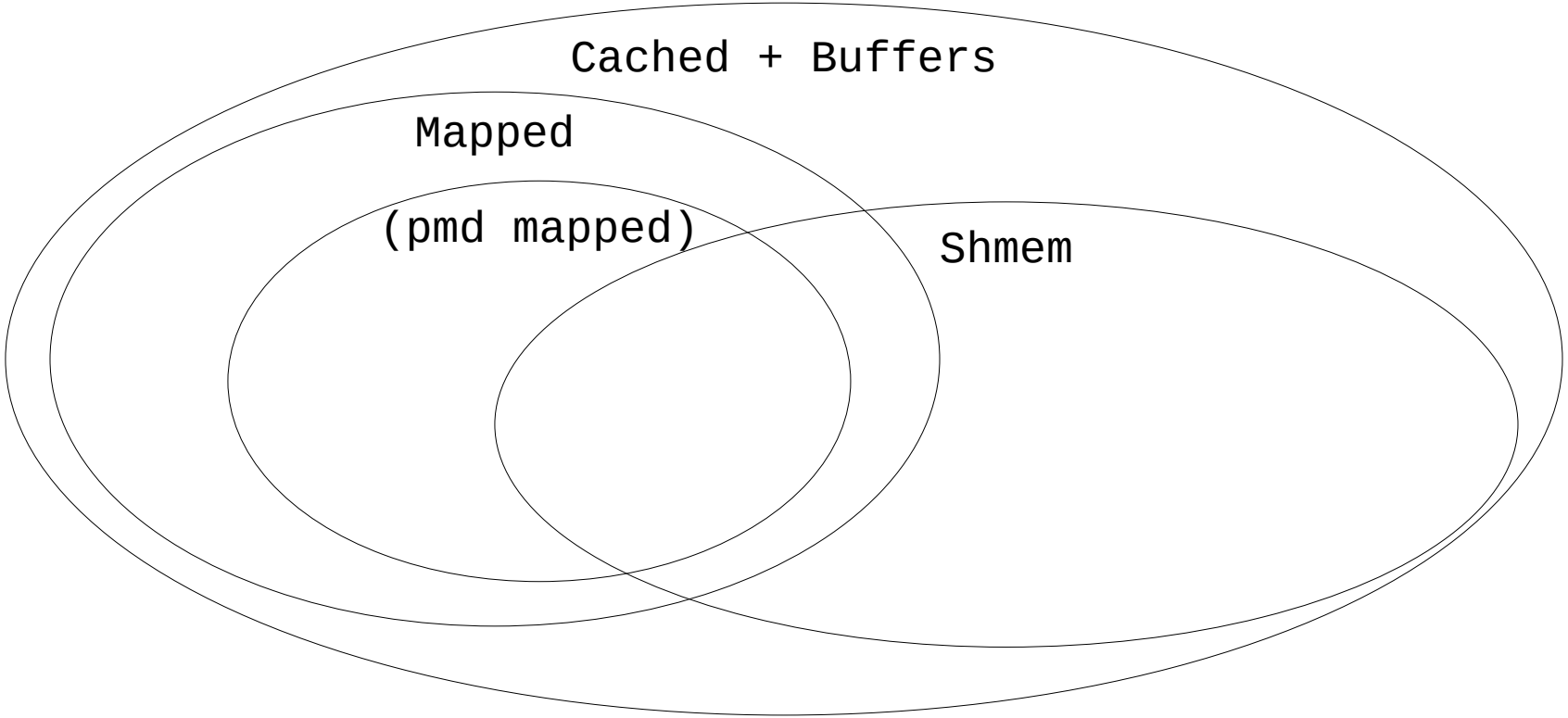
Page cache counters in detail – (Pmd)Mapped



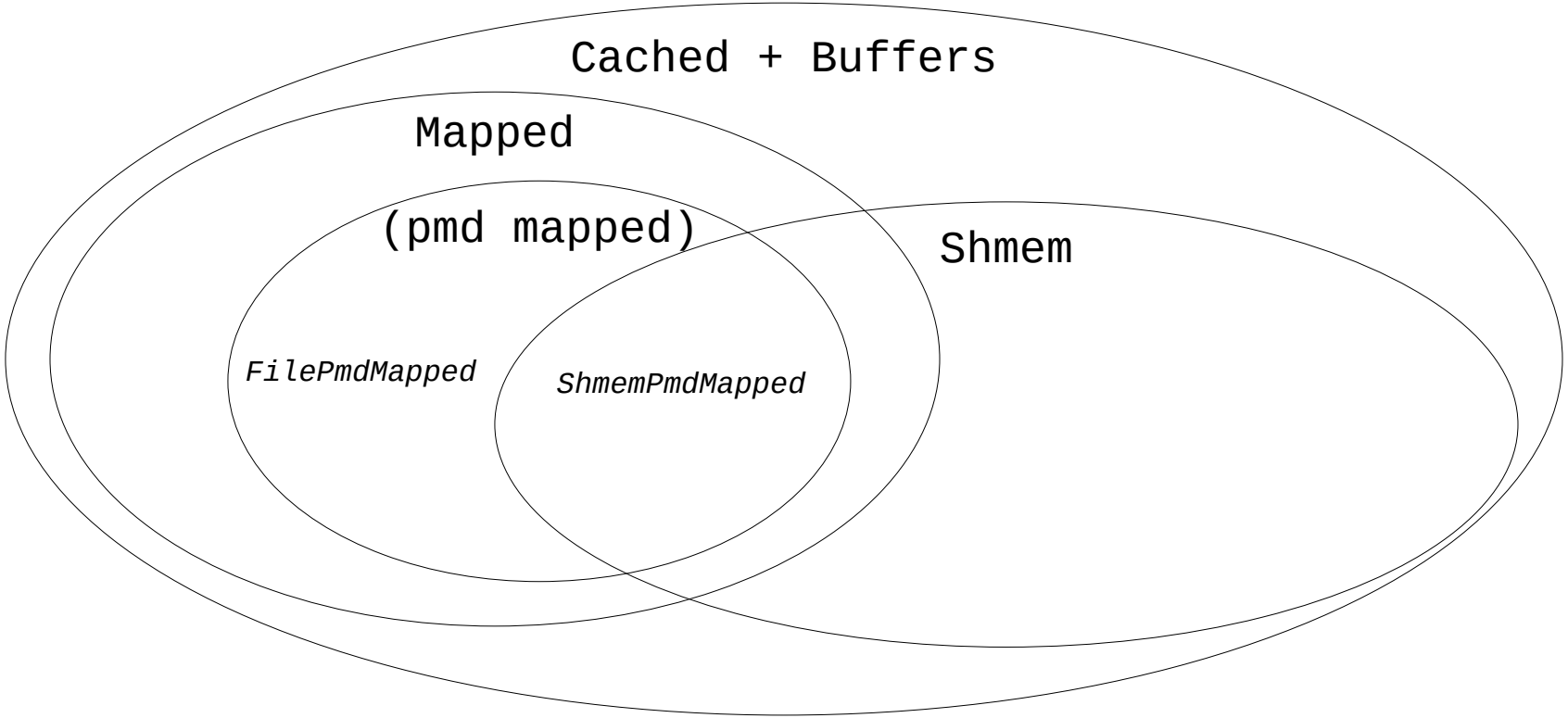
Page cache counters in detail – (Pmd)Mapped



Page cache counters in detail – (Pmd)Mapped



Page cache counters in detail – (Pmd)Mapped



Active and Inactive

- Userspace pages are sorted by their “hotness” on least recently used (LRU) lists
  - divided to anon and file due to their different characteristics
  - divided to active and inactive parts for reclaim optimization purposes
  - pages near the tail of the inactive list are on their way to destruction by kswapd or direct reclaim
- The counters are a redundant way to see how much memory is used by the page cache and anonymous pages
  - With a twist: *Shmem* is in page cache (*Cached*) but on *anon* LRU (because it’s swap backed)
  - Unevictable/Mlocked may account both page cache and anonymous pages
- `AnonPages + Shmem = 8922944 + 611016 = 9533960 kB`
  `Active(anon) + Inactive(anon) = 9589756 + 8120 = 9597876 kB`
- `Cached + Buffers – Shmem = 12902952 + 258820 – 611016 = 12550756 kB`
  `Active(file) + Inactive(file) = 5862552 + 6688184 = 12550736 kB`

Legend: Kernel Userspace *Redundant* Not consumed memory **Sums to MemTotal**

MemTotal:	32125696 kB	SwapFree:	2513056 kB	NFS_Unstable:	0 kB
MemFree:	**8627220 kB**	**Zswap:**	**0 kB**	**Bounce:**	**0 kB**
MemAvailable:	21107232 kB	Zswapped:	0 kB	**WritebackTmp:**	**0 kB**
Buffers:	**258820 kB**	*Dirty:*	*844 kB*	CommitLimit:	18575904 kB
Cached:	**12902952 kB**	*Writeback:*	*196 kB*	Committed_AS:	21267524 kB
SwapCached:	*0 kB*	**AnonPages:**	**8922944 kB**	VmallocTotal:	34359738367 kB
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	**VmallocUsed:**	**84632 kB**
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	VmallocChunk:	0 kB
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	**Percpu:**	**7680 kB**
Inactive(anon):	*9589756 kB*	**Slab:**	**613748 kB**	*AnonHugePages:*	*1363968 kB*
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	*ShmemHugePages:*	*0 kB*
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	*ShmemPmdMapped:*	*0 kB*
Unevictable:	*80 kB*	**KernelStack:**	**35568 kB**	*FileHugePages:*	*1062912 kB*
Mlocked:	*80 kB*	**PageTables:**	**82500 kB**	*FilePmdMapped:*	*684032 kB*
SwapTotal:	2513056 kB	**SecPageTables:**	**3248 kB**	**Hugetlb:**	**0 kB**

So which fields sum to MemTotal?

- `MemFree + Buffers + Cached + Zswap + AnonPages + Slab + KernelStack + PageTables + SecPageTables + Bounce + WritebackTmp + VmallocUsed + Percpu + Hugetlb`
  `= 31539312 kB`
- `MemTotal = 32125696 kB`
  - `572 MB` not accounted for by any `/proc/meminfo` counter – What !
  - `Unaccounted` field proposed as RFC in 2016 – mixed feedback
- What you say about MemAvailable then?
  - An estimate of how much is free or reclaimable (except anonymous)
  - `MemFree` (minus some reserve)
    `+ Active(file)` `+ Inactive(file)` (minus some reserve)
    `+ KReclaimable` (minus some reserve)
    `= 21568012 kB` (reserves not subtracted)
  - `MemAvailable: 21107232 kB`

Determining process memory usage – ps?

- ps, top, htop… all have columns for process memory usage
  - VSZ or VIRT (virtual memory size)
    - sizes of `mmap()` areas (VMA's) only
    - may count towards `Committed_AS`, but does not say how much memory is actually occupied
  - RSS or RES (resident set size)
    - sum of the populated parts (actually backed by memory) of VSZ
    - better, but also not perfect
  - SHR, CODE, DATA… and other subsets of RSS
    - More specific use for some of the resident memory

How can RSS be misleading ?

- Demo: an simple program named zig
  - Create a 1GB area with `mmap(MAP_ANONYMOUS | MAP_PRIVATE)`
  - Fault in all the pages in the area by writing a byte in a loop with 4kB step
  - `fork()` 10 children processes of the main process
  - Rewrite the area in each child
  - Observe system memory usage and RSS of the test processes in each step
    - `free -m ; ps -C zig u`

How can RSS be misleading ?

- Create a 1GB area with `mmap(MAP_ANONYMOUS | MAP_PRIVATE)`
  - ```
                total        used        free      shared  buff/cache   available
    Mem:        31372       12229       10060         647       10179       19143

      PID    %CPU    %MEM     VSZ     RSS    COMMAND
    78893     0.0     0.0 1051220    1504    ./zig
    ```
  - VSZ is 1GB but RSS not yet
- Fault in all the pages in the area by writing a byte in a loop with 4kB step
  - ```
                total        used        free      shared  buff/cache   available
    Mem:        31372       13360        8926         647       10183       18012

      PID    %CPU    %MEM     VSZ     RSS    COMMAND
    78893     0.1     3.2 1051220 1050200    ./zig
    ```
  - RSS is now also 1GB, used memory increased by 1GB

How can RSS be misleading ?

- Fault in all the pages in the area by writing a byte in a loop with 4kB step
  - ```
                total        used        free      shared  buff/cache   available
    Mem:        31372       13360        8926         647       10183       18012

      PID    %CPU    %MEM      VSZ      RSS    COMMAND
    78893     0.1     3.2  1051220  1050200    ./zig
    ```
- `fork()` 10 children processes of the main process
  - ```
                total        used        free      shared  buff/cache   available
    Mem:        31372       13459        8823         653       10192       17912

      PID    %CPU    %MEM      VSZ      RSS    COMMAND
    78893     0.0     3.2  1051220  1050328    ./zig
    79302     0.0     3.2  1051220  1049144    ./zig
    …
    79311     0.0     3.2  1051220  1049108    ./zig
    ```
  - Sum of RSS is 11GB but used memory almost unchanged – Ha ha ha ha …

How can RSS be misleading ?

- `fork()` 10 children processes of the main process
  - ```
                total        used        free      shared  buff/cache   available
    Mem:        31372       13459        8823         653       10192       17912

      PID    %CPU    %MEM       VSZ       RSS    COMMAND
    78893     0.0     3.2   1051220   1050328    ./zig
    79302     0.0     3.2   1051220   1049144    ./zig
    …
    79311     0.0     3.2   1051220   1049108    ./zig
    ```
- Rewrite the area in each child
  - ```
                total        used        free      shared  buff/cache   available
    Mem:        31372       23741         226         647        8501        7630

      PID    %CPU    %MEM       VSZ       RSS    COMMAND
    78893     0.1     3.2   1051220   1050328   ./zig
    79302     0.2     3.2   1051220   1049144   ./zig
    …
    79311     0.0     3.2   1051220   1049532   ./zig
    ```
  - Used memory increased by 10GB, RSS unchanged

You know what you doing. (in the kernel)

- Kernel has per-process (mm) RSS counters of several (mutually exclusive) types
  - resident file pages, anonymous pages, shmem pages, and also swap entries
  - increased and decreased as respective pages (swap entries) get mapped or unmapped in the page tables
  - exported by e.g. `/proc/$PID/status` along with others
- `mmap()` without `MAP_POPULATE` doesn't populate the page tables – no RSS increase
- Writing with 4kB step populates page tables with anonymous pages – RSS increased
- `fork()` copies both the VMA's and page tables of the parent to the child (with exceptions)
  - the RSS counters will have almost the same values
  - but the pages are not duplicated, instead they become transparently shared
- Only on child writes, new pages are allocated by the copy on write (CoW) mechanism
  - No change to the RSS, but memory usage increased

Why is there no accurate counter ?

- Could we maintain a per-process counter where memory usage of shared pages would be weighted by how many processes share them?
  - Defined as PSS (proportional set size)
  - Sum over all processes would match the total used memory
  - It would be expensive – i.e. a CoW in one process would have to increment counters in other processes
  - RSS is imperfect but simple and efficient to maintain :)
- BTW, how does OOM killer determine which process have no chance to survive make your time?
  - Also RSS (all types of entries, including swap), plus memory occupied by task's page tables
  - With almost all pages shared, killing one of our test processes would actually free nothing…
    - We would have to take off every 'ZIG'!! to free any memory
  - Better metric – USS (unique/unshared set size) – count pages not shared with anyone else
    - Not maintained as that would be similarly expensive to maintaining PSS

Pay the price for accuracy on-demand?

- PSS and USS can be provided by the kernel, but it's a snapshot calculated on the fly
  - `/proc/$PID/smaps(_rollup)`
  - Has to walk the page tables, and inspect individual mapped pages' mapcount and other attributes
  - Holds `mmap_lock` for read, which can block the inspected process taking `mmap_lock` for write, and result in priority inversion
    - Low priority monitoring task might be blocking high priority workload
    - There is an an ongoing work to fix this

Pay the price for accuracy on-demand?

- Example (some fields omitted) for our demo program after the fork()s:
  - ```
    > cat /proc/69788/smaps_rollup
    00400000-7ffdb60d9000 ---p 00000000 00:00 0                              [rollup]
    Rss:                1050416 kB
    Pss:                  95386 kB
    Pss_Dirty:            95374 kB
    Pss_Anon:             95365 kB
    Pss_File:                12 kB
    Pss_Shmem:                8 kB
    Shared_Clean:          1700 kB
    Shared_Dirty:       1048676 kB
    Private_Clean:            0 kB
    Private_Dirty:           40 kB
    Referenced:         1050416 kB
    Anonymous:          1048704 kB
    AnonHugePages:      1048576 kB
    ShmemPmdMapped:           0 kB
    FilePmdMapped:            0 kB
    Swap:                     0 kB
    SwapPss:                  0 kB
    Locked:                   0 kB
    ```
    ```

Pay the price for accuracy on-demand?

- Example (some fields omitted) for our demo program after the children writes
  - ```
    > cat /proc/69788/smaps_rollup
    00400000-7ffdb60d9000 ---p 00000000 00:00 0                              [rollup]
    Rss:             1050204 kB
    Pss:             1048629 kB
    Pss_Dirty:       1048617 kB
    Pss_Anon:        1048616 kB
    Pss_File:             11 kB
    Pss_Shmem:             1 kB
    Shared_Clean:       1488 kB
    Shared_Dirty:        108 kB
    Private_Clean:         0 kB
    Private_Dirty:   1048608 kB
    Referenced:      1050204 kB
    Anonymous:       1048704 kB
    AnonHugePages:     16384 kB
    ShmemPmdMapped:        0 kB
    FilePmdMapped:         0 kB
    Swap:                  0 kB
    SwapPss:               0 kB
    Locked:                0 kB
    ```
    ```

Pay the price for accuracy on-demand?

- Example (some fields omitted) for our demo program after the children writes
  - ```
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    00400000-7ffdb60d9000 ---p 00000000 00:00 0                              [rollup]
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    Pss:                1048629 kB
    Pss_Dirty:          1048617 kB
    Pss_Anon:           1048616 kB
    Pss_File:                11 kB
    Pss_Shmem:                1 kB
    Shared_Clean:          1488 kB
    Shared_Dirty:           108 kB
    Private_Clean:            0 kB
    Private_Dirty:      1048608 kB
    Referenced:         1050204 kB
    Anonymous:          1048704 kB
    AnonHugePages:        16384 kB
    ShmemPmdMapped:           0 kB
    FilePmdMapped:            0 kB
    Swap:                     0 kB
    SwapPss:                  0 kB
    Locked:                   0 kB
    ```

(dirty) pages that only this process is mapping
~ USS
    ```

More user friendly usage – the smem tool

- The demo program after populating the area:
  -
```
   PID   Command        Swap      USS      PSS      RSS
 72922   ./zig             0  1048728  1048756  1050288
```
- After fork()s
  -
```
   PID   Command        Swap      USS      PSS      RSS
 73362   ./zig             0       28    95369  1049404
 …
 72922   ./zig             0       36    95388  1050416
```
- After children writes
  -
```
   PID   Command        Swap      USS      PSS      RSS
 73362   ./zig             0  1048604  1048618  1049412
 …
 73371   ./zig             0  1048620  1048636  1049732
```

How to debug *missing* kernel memory?

- Unaccounted amount too large or growing over time?
  - Somebody set up us the leak. But who?
- `CONFIG_PAGE_OWNER=y` and boot with `page_owner=on`
  - Some CPU and memory overhead, but none unless enabled on boot
  - `/sys/kernel/debug/page_owner` – per-page allocation stacks and other info, requires postprocessing to aggregate
  - `/sys/kernel/debug/page_owner_stacks/show_stacks` – new aggregate output
    - Not sorted, but `echo X > .../count_threshold` can limit the output
  - freeing stacks also collected, can be dumped in case some page corruption is detected (i.e. debug_pagealloc)

How to debug *missing* kernel memory?

- Example `/sys/kernel/debug/page_owner_stacks/show_stacks`
  - ```
    get_page_from_freelist+0x1004/0x1210
    __alloc_pages_noprof+0x18f/0x10b0
    new_slab+0xc7/0x3c0
    ___slab_alloc+0x75f/0xbe0
    kmem_cache_alloc_lru_noprof+0x1e8/0x210
    fuse_alloc_inode+0x26/0x110
    alloc_inode+0x1b/0x60
    iget5_locked+0x97/0xe0
    fuse_iget+0x63/0x230
    fuse_lookup_name+0x16d/0x250
    fuse_lookup+0x7f/0x1a0
    __lookup_slow+0x89/0x130
    lookup_one_unlocked+0x93/0xb0
    ovl_lookup_single+0x7b/0x370
    ovl_lookup_layer+0x12d/0x150
    ovl_lookup+0x72e/0xd80
    nr_base_pages: 18880
    ```
    ```

How to debug *missing* kernel memory?

- What if we know it's a particular slab cache? (from `/proc/slabinfo`)
  - For example, `kmalloc-*` caches have many different users
  - Boot with a `slab_debug` option, e.g. `slab_debug=U,kmalloc-*`
  - `cat /sys/kernel/debug/slab/kmalloc-64/alloc_traces`
    - No minimum threshold that could be set, but it is sorted from the largest user
    - ```
          182 scsi_dev_info_list_add_keyed+0x99/0x1a0 waste=1456/8
      age=98885/98885/98885 pid=1 cpus=7
              scsi_dev_info_list_add_keyed+0x99/0x1a0
              scsi_init_devinfo+0x5b/0xa0
              init_scsi+0x1a/0x90
              do_one_initcall+0x5b/0x230
              kernel_init_freeable+0x1d3/0x220
              kernel_init+0x1a/0x130
              ret_from_fork+0x34/0x50
              ret_from_fork_asm+0x1a/0x30
      ```
      ```

How to debug *missing* kernel memory?

- A new possible alternative to both of page_owner and slab_debug mechanisms:
- `CONFIG_MEM_ALLOC_PROFILING`, `sysctl.vm.mem_profiling=0|1|never`
  - Can be enabled/disabled after boot, but the memory overhead paid on boot
  - Not capturing full stacks, but callsites (file, line, function)
    - If some helper is used by different users, it needs to be wrapped in a macro to distinguish them
- # sort -g /proc/allocinfo | tail -n 11 | numfmt --to=iec
  -
```
1.9M     9923 fs/dcache.c:1636 func:__d_alloc
2.1M      513 mm/execmem.c:31 func:__execmem_alloc
2.1M      655 kernel/fork.c:313 func:alloc_thread_stack_node
2.1M      528 drivers/virtio/virtio_ring.c:319 func:vring_alloc_queue
2.5M      616 mm/mm_init.c:2428 func:alloc_large_system_hash
3.7M      925 mm/shmem.c:1722 func:shmem_alloc_folio
4.5M     1134 mm/memory.c:1062 func:folio_prealloc
8.0M     2048 drivers/char/virtio_console.c:442 func:alloc_buf
9.4M     2384 mm/memory.c:1060 func:folio_prealloc
 28M     2557 mm/slub.c:2325 func:alloc_slab_page
 45M    11299 mm/readahead.c:248 func:page_cache_ra_unbounded
```

How to debug *missing* kernel memory?

- What if I don’t want to reboot or even rebuild the kernel?
- Tracing! Capture memory allocation and freeing events.
  - Store stacktrace of the allocation events and allocated page address
  - Drop the stored event when freeing the same address
    - Postprocessing of the trace, or in the kernel itself by eBPF
  - At any point, report outstanding allocations (not yet freed) - It's you !!
    - If the memory leak is still occurring, it will eventually dominate the report
- Example: memleak.py in bcc-tools

Final PSA: non-zero swap usage is not a bug

- Consider a system that has enough memory to fit its workload and some page cache
- An overnight activity (backup, virus scanning) results in accessing many more files
- No swap usage whatsoever with older kernels
- Suddenly, some swap becomes used after a kernel/distro upgrade – why?
  - Same workload, same amount of memory?
- The answer: kernel got better at tracking unused file vs anon memory around v5.9
  - Workingset tracking expanded to anon memory
  - Previously reclaim very biased towards page cache, reluctant to swap out anon
  - Now able to detect anonymous page not touched *at all* by their processes
    - Vs file pages touched at least once a day due to the overnight activity
    - Swapping out is the correct thing to do! More memory for page cache, for great justice!



SUSE

Thank you

Questions?

